



MAHMUT ESAT KOLAY

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EDUCATION



Selcuk University Bachelor of Computer Engineering

Sep 2023 - Jun 2027

- Studied fundamental topics: algorithms, data structures, object-oriented programming, databases



Mehmet Tuza Pakpen Technical High School

Sep 2019 - Jun 2023

- Studied core topics: microcontroller programming, circuit design, PLC systems, and industrial sensors

PROJECTS

Microsoft Foundry Local — Industrial Automation Copilot

Tech: Python, FastAPI, RAG, LLMs, BM25, Vector Search, Cross-Encoder

- Developed a fully local industrial RAG assistant supporting technical PDF, DOCX, and XLSX documents with hybrid retrieval and citation-based responses.
- Implemented query rewriting, Reciprocal Rank Fusion, Cross-Encoder re-ranking, Self-RAG grading, and entity-aware retrieval for more reliable technical Q&A.

Home Robotics with MuJoCo & ROS 2 — METU ROMER Research

Tech: Python, ROS 2, MuJoCo, Franka Panda, VLA, Robotics, Simulation

- Developed a dual-arm home robotics framework integrating ROS 2, MuJoCo, and Vision-Language-Action workflows for robotic manipulation.
- Built a lightweight dual Franka Panda simulation environment with structured control architecture, manipulation pipelines, and reproducible task execution.

Explainable Aerospace Battery RUL Prediction

Tech: Python, LSTM, Attention, Explainable AI

- Developed an explainable LSTM + attention-based RUL prediction model for aerospace Li-Ion batteries across multiple thermal conditions.
- Achieved ~84% RMSE reduction in critical cases and was selected for T3 Foundation Young Researcher Support.

TÜBİTAK 2209-A — Deep Learning-Based Oilseed Classification

Tech: Python, Deep Learning, Computer Vision, Image Processing

- Developed an image processing and deep learning pipeline for detection and six-class classification of oilseeds.
- Evaluated multiple deep learning-based approaches to improve classification performance on a 10K+ image dataset.

AI-Driven Inventory Optimization & Demand Forecasting

Tech: Python, Pandas, XGBoost, Time-Series Analysis

- Developed an end-to-end FMCG inventory optimization framework using XGBoost, time-series features, and dynamic safety-stock modeling.
- Reduced simulated inventory costs by 32.6% while maintaining a 95% service level under demand volatility.

ACHIEVEMENTS & COMPETITIONS

Unilever Factory Experience @HPC — 1st Place

- Ranked 1st in the Factory Experience program, applying data-driven problem solving to real-world manufacturing challenges.

EKA — Smart Vertical Transport & Robotic Construction Competition

- Top 10 finalist among 60 teams in a robotic construction competition.

Unilever Chain Reaction @ U-House

- Participated in a competitive supply chain simulation, applying data-driven optimization and decision-making strategies.

BTK Ankara AI Hackathon @ Ankara Teknopark

- Built a functional AI prototype within 30 hours under hackathon constraints.

TECHNICAL SKILLS

AI & Machine Learning: LLMs, RAG, Deep Learning, Computer Vision, LSTM, Attention, XGBoost

Robotics & Simulation: ROS 2, MuJoCo, Gazebo, Isaac Sim, Franka Panda, KUKA, KRL, PLC

Retrieval & Backend: BM25, Vector Search, Cross-Encoder, FastAPI, SQL

Programming Languages: Python, C++, C#, Java

DevOps & Tools: Git, Docker, Linux, Pandas

WORK EXPERIENCE



METU ROMER - Visiting Research Intern

Aug 2026 - Sep 2026

- Analyzed ROS and Isaac Sim workflows for home robotics, exploring Vision-Language-Action (VLA) models for manipulation tasks.
- Developed a lightweight MuJoCo simulation environment on Apple Silicon, integrating ROS 2 control architectures for dual Franka Panda arms.



Microsoft Türkiye - Summer AI Intern

June 2026 - Aug 2026

- Designed and developed a local AI solution for industrial automation using Microsoft Foundry Local and RAG architectures.
- Built a modular retrieval and generation pipeline combining hybrid search, re-ranking, and relevance evaluation.



FAB Tech. Inc - Product Management Intern

Sep 2025 - Nov 2025

- Completed an AI & Startup Strategy Fellowship, applying Founder and Project Mindset to develop ideas from 0 to 1.
- Built workflow automations with UiPath and used Google AI tools to create generative AI-driven business solutions.



OSR Robotics - Robotics Intern

June 2025 - July 2025

- Designed and implemented ML-driven fault detection algorithms for industrial robotic systems
- Programmed KUKA robots and analyzed motion through kinematics and dynamic modeling



Komalp - Industrial Automation Intern

June 2023 - July 2023

- Participated in the installation, maintenance, and troubleshooting processes of factory electrical systems.
- Conducted assembly, fault detection, and monitoring operations for power factor correction (PFC) panels.



Devpak Machine - Industrial Automation Intern

June 2022 - Oct 2022

- Assisted in PLC programming, control panel design, and machine wiring for industrial automation systems
- Observed and analyzed existing automation systems to identify improvements in operational efficiency

CERTIFICATES



TEI Aviation Engines School



Unilever Supply Chain + Chain Reaction



Aspire Institute - Aspire Leaders Program



Deloitte Australia - Data Analytics



Fundamentals of Deep Learning



Academy Of Responsible Development Erasmus+



Intermediate Machine Learning

REFERENCES

Barbaros Günay

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